

# Software Requirements Specification

Hyperlocal Courier Dispatch & Tracking Engine  
Problem Statement #23 | Smart Cities, Transport & Logistics  
Version 1.0

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## 1. Introduction

### 1.1 Purpose

This document specifies the functional and non-functional requirements of the Hyperlocal Courier Dispatch & Tracking Engine. It is intended for the developers, testers and evaluators of the system and forms the basis for design, implementation and testing.

### 1.2 Scope

The system is an on-demand package delivery platform. It lets a sender request a local parcel pickup, assigns the request to the nearest active delivery rider, generates and optimizes the rider's route when several stops are assigned, lets the sender track the delivery, and requires OTP verification at the destination before a delivery is marked complete.

Features not covered by this specification, such as payments, billing, ratings and customer support, are out of scope.

### 1.3 Definitions and abbreviations

| Term                  | Meaning                                                                          |
|-----------------------|----------------------------------------------------------------------------------|
| Sender Client         | The user who creates a courier request and tracks the delivery.                  |
| Delivery Rider        | The courier who receives dispatch requests, collects the parcel and delivers it. |
| Routing / GPS Service | External service that provides maps, route computation and rider location data.  |
| Active rider          | A rider who is online and available to receive a dispatch request.               |
| Courier request       | A request containing the pickup location, delivery location and parcel details.  |
| OTP                   | One-time password used to confirm that the parcel reached the correct recipient. |
| FR / NFR              | Functional Requirement / Non-Functional Requirement.                             |
| SRS / RTM             | Software Requirements Specification / Requirements Traceability Matrix.          |

### 1.4 References

- Lab 1 Problem Statement #23 - Hyperlocal Courier Dispatch & Tracking Engine
- Lab1\_Requirements\_Table (folder 1-RE) and Lab1\_RTM\_Table (folder 1-RE)
- UML use-case diagram and Use-Case Flow Specification (folder 2-Architectural\_Diagram)
- Jira projects: Kanban\_BPS#23, Scrum\_BPS#23 and Bug\_Tracking\_BPS#23 (folder 3-Project\_Creation\_Screenshots)

## 2. Overall Description

### 2.1 Product perspective

The platform has a Sender Client interface, a Delivery Rider interface and a back-end dispatch engine. The dispatch engine depends on an external Routing / GPS Service for rider locations, route computation and live tracking data.

### 2.2 Product functions

- Create a courier request with pickup, delivery and parcel details (UC-01).
- Match the request with the nearest active rider within 3 km (UC-02).
- Let the rider receive, view, accept or reject the dispatch (UC-03).
- Let the sender track the delivery with live location and status (UC-04, UC-05).
- Plan the delivery route and optimize multi-stop routes (UC-06, UC-07).
- Complete the delivery after OTP verification at the destination (UC-08, UC-09).

### 2.3 User classes and actors

| Actor                 | Type                  | Role in the system                                                                             |
|-----------------------|-----------------------|------------------------------------------------------------------------------------------------|
| Sender Client         | Primary               | Creates courier requests and tracks deliveries.                                                |
| Delivery Rider        | Primary               | Receives and responds to dispatches, follows the route, updates status and completes delivery. |
| Routing / GPS Service | Supporting (external) | Supplies location data, route computation and tracking information.                            |

### 2.4 Constraints

- Rider matching is limited to a 3 km radius around the pickup location.
- A delivery can only be marked as delivered after a correct OTP is verified.
- Live GPS and status updates must reach the Sender Client in under 2 seconds.
- The system must reach at least 99.5% availability during normal operation.

### 2.5 Assumptions and dependencies

- Active riders share their location with the system while they are available.
- The external Routing / GPS Service is reachable and returns location and route data.
- The sender enters valid pickup and delivery locations.
- An OTP is generated for each delivery and is available to the recipient to give to the rider at the destination.

## 3. Specific Requirements

### 3.1 Functional requirements

| ID     | Description                                                                                                                                 | Priority | Use case     | Acceptance criteria                                                                                                                                                 |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-001 | The system shall match a courier request with the nearest active delivery rider within a 3 km radius.                                       | High     | UC-01, UC-02 | Pass: Nearest active rider within 3 km receives the request. Fail: An offline or farther rider is assigned when a nearer active rider is available.                 |
| FR-002 | The system shall allow the Sender Client to create a courier request by entering the pickup location, delivery location and parcel details. | High     | UC-01        | Pass: Request is created only after all required details are entered and a request ID is generated. Fail: Request is created with required details missing.         |
| FR-003 | The system shall allow the Delivery Rider to accept or reject a courier request assigned to them.                                           | High     | UC-03        | Pass: Accepting changes the status to Accepted and notifies the sender; rejecting starts reassignment. Fail: A rejected request remains assigned to the same rider. |
| FR-004 | The system shall generate a delivery route for the rider and optimize the order of stops when multiple deliveries are assigned.             | High     | UC-06, UC-07 | Pass: The route includes all assigned stops in an ordered route. Fail: Any assigned pickup or delivery stop is missing.                                             |
| FR-005 | The system shall verify an OTP at the destination before the courier can be marked as delivered.                                            | High     | UC-08, UC-09 | Pass: A correct OTP changes the status to Delivered. Fail: An incorrect OTP allows the delivery to be completed.                                                    |

### 3.2 Non-functional requirements

| ID      | Type                       | Description                                                                                        | Priority | Use case     | Acceptance criteria                                                                                                                                     |
|---------|----------------------------|----------------------------------------------------------------------------------------------------|----------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-001 | Performance & Security     | Delivery status and live GPS updates from riders shall reach the Sender Client in under 2 seconds. | High     | UC-04, UC-05 | Pass: GPS and status updates reach the Sender Client in under 2 seconds during peak-load testing. Fail: Updates take 2 seconds or more or are insecure. |
| NFR-002 | Availability / Reliability | The system shall maintain at least 99.5% availability during normal operation.                     | Medium   | All          | Pass: Monthly system availability is at least 99.5%. Fail: Monthly availability falls below 99.5%.                                                      |

### 3.3 Use-case summary

| Use case | Name                       | Actor(s)                              | Description                                                                                                             |
|----------|----------------------------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| UC-01    | Create Courier Request     | Sender Client                         | Sender enters pickup, delivery and parcel details; the system creates the request and starts matching (includes UC-02). |
| UC-02    | Match Nearest Active Rider | System; Routing / GPS Service         | Finds active riders within 3 km of the pickup location and selects the nearest one.                                     |
| UC-03    | Receive / View Dispatch    | Delivery Rider                        | Rider receives the dispatch with the pickup location and accepts or rejects it.                                         |
| UC-04    | Track Delivery             | Sender Client; Routing / GPS Service  | Sender views live location and status of the delivery.                                                                  |
| UC-05    | Update Delivery Status     | Delivery Rider                        | Rider updates the delivery status as the job progresses.                                                                |
| UC-06    | Plan Delivery Route        | Delivery Rider                        | System generates the route covering all assigned stops.                                                                 |
| UC-07    | Optimize Multi-Stop Route  | Delivery Rider; Routing / GPS Service | Optional extension of UC-06 that orders multiple stops efficiently.                                                     |
| UC-08    | Complete Delivery          | Delivery Rider                        | Rider completes the delivery at the destination (includes UC-09).                                                       |
| UC-09    | Verify Destination OTP     | Delivery Rider                        | System checks the OTP entered at the destination before marking the parcel delivered.                                   |

### 3.4 External interface requirements

- **User interfaces:** a Sender Client interface for creating requests and tracking, and a Delivery Rider interface for dispatches, routes, status updates and OTP entry.
- **Software interface:** the external Routing / GPS Service, used for rider location, route computation and tracking data.
- **Notifications:** the rider receives a dispatch notification with the pickup location; the sender is notified when the request is accepted and when its status changes.

## 4. Requirements Traceability

Every requirement above is traced to its use case, Jira epic, Kanban story, Scrum story and bug in the Requirements Traceability Matrix (Lab1\_RTM\_Table, folder 1-RE).